Abstract

Abstract I

Numbered-table references point to the canonical manuscript; they are not standalone local deck numbers.

DuckRabbit is typed, deterministic research software for constructing reproducible visual, auditory, temporal, and audiovisual stimulus families. It is published and citable at the DuckRabbit public GitHub repository and released under the MIT License.

Its basic unit is an immutable request containing an illusion identifier, validated parameters, a seed, and an encoding specification. The request yields a canonical artifact, objective media measurements, and a versioned provenance manifest before any delivery codec is selected. This separation makes the stimulus a testable computational object while reserving claims about perception for controlled observer protocols.

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The release situates its engineering choices within a deliberately broad historical foundation spanning Greek, Arabic/Islamicate, Chinese, and early-modern work on optics, visual inference, and cross-sensory knowledge, alongside later psychophysics and illusion research. These traditions motivate questions about construction, observation, and evidence; they are contextual precedents, not evidence that a generated file reproduces an ancient, early-modern, or clinical observation.

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Version 0.6.0 contains 17 implemented generators and 18 catalog entries, supported by 36 source records and 18 evidence records in a checked-in audit snapshot. The package generates 15 publication figures and 10 machine-derived tables from the live registry. It also provides a typed observer-study harness and a transparent synthetic diagnostic: a hand-specified feature observer with serialized weights, temperature, analytic calibration, and `human_data=false`. No participant data are bundled; synthetic model output is explicitly nonhuman.

Abstract I

DuckRabbit verifies physical stimulus properties, media round trips, hashes, timing, and declared provenance. It does not infer a universal percept, effect size, or cross-device perceptual equivalence from a generated artifact. Its contribution is a reproducibility and epistemic boundary: source-backed family descriptions, deterministic media facts, and future observer hypotheses remain different typed records rather than being collapsed into one claim. DuckRabbit is software for reproducible stimulus construction and audit, not a claim that a generated file alone produces a universal perceptual effect.